Transformations

Name _____ Date _____ No calculator Version 1 Scan to practise this topic online

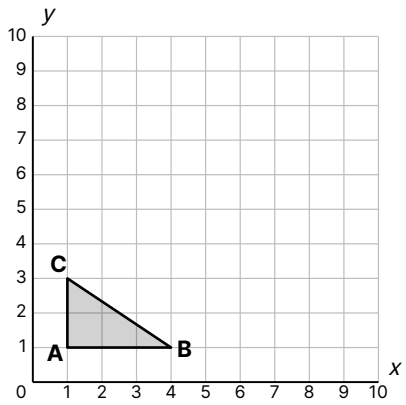
KEY WORDS translation *(nekehanga)* · reflection *(whakaata)* · rotation *(hurihanga)* · mirror line *(rārangi whakaata)*

A Translations ●○○

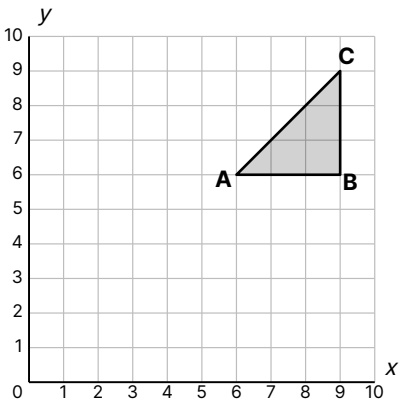
Remember: A translation slides a shape. Every corner moves the same distance in the same direction. Label the corners of the image A′, B′, C′.

1. Draw the image of each shape.

a) Translate the shape 5 units right and 4 units up.

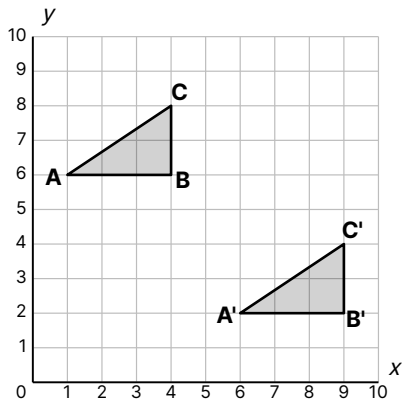


b) Translate the shape 5 units left and 3 units down.



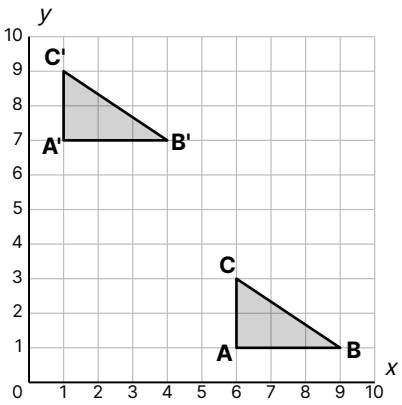
2. Shape ABC has been translated to A′B′C′. Describe the translation. Say how many units it has moved right or left, and up or down.

a)



_____

b)



_____

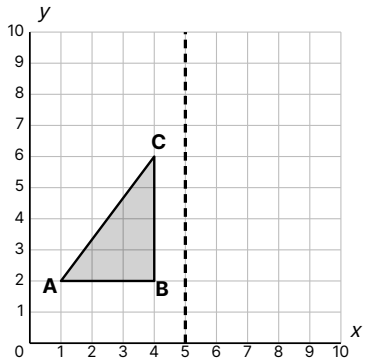
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B Reflections ●●○

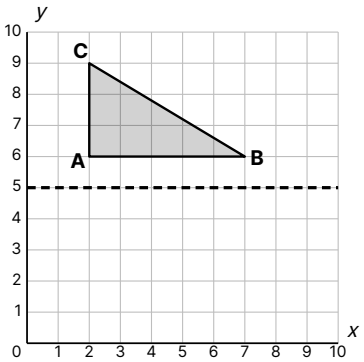
Remember: A reflection flips a shape over a mirror line. Each corner and its image are the same distance from the mirror line, on opposite sides of it.

3. Draw the image of each shape.

a) Reflect the shape in the dashed mirror line.



b) Reflect the shape in the dashed mirror line.

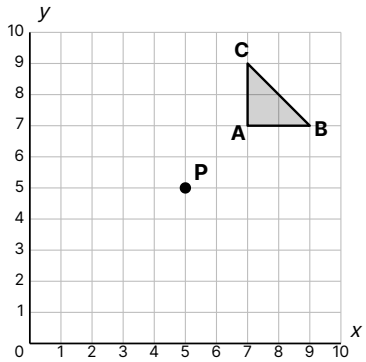


C Rotations ●●●

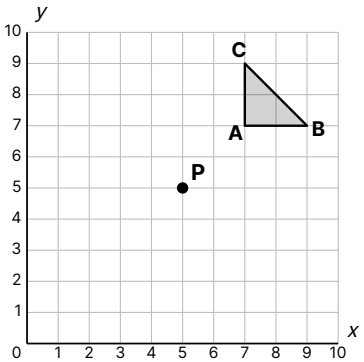
Remember: A rotation turns a shape about a point. Tracing paper helps: hold the point P still and turn the paper.

4. Draw the image of each shape.

a) Rotate the shape a half turn (180°) about the point P.



b) Rotate the shape a quarter turn (90°) clockwise about the point P.



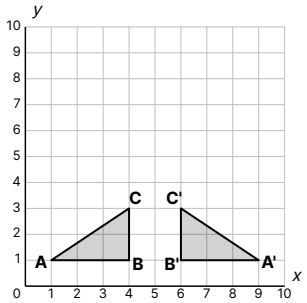
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D Naming the transformation ●●○

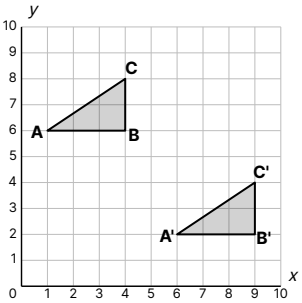
Remember: Translation: a slide. Reflection: a flip. Rotation: a turn.

5. Shape ABC has been moved to A′B′C′. Is each move a translation, a reflection or a rotation?

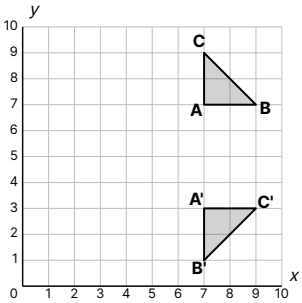
a)



b)



c)



6. Which kind of transformation is being described?

a) A shape slides 4 units to the left. _____

b) A shape is flipped over a mirror line. _____

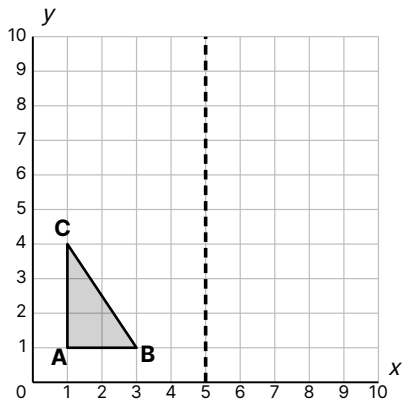
c) A shape makes a quarter turn about one of its corners. _____

E Two-step transformations ●●●

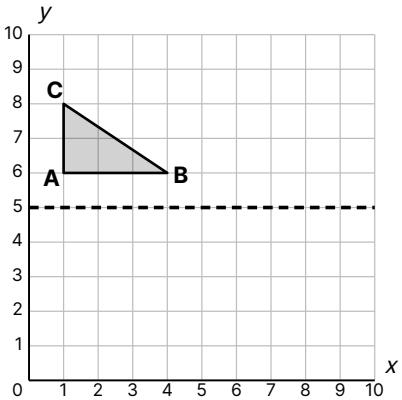
Watch out: Do one step at a time. Draw the shape after the first step lightly, then move that new shape for the second step.

7. Draw the final image of each shape.

a) Translate the shape 1 unit right and 4 units up. Then reflect the new shape in the dashed mirror line.



b) Reflect the shape in the dashed mirror line. Then translate the new shape 5 units right.



★ Extension

8. Without drawing, write the coordinates of the image of the point $(3,\ 4)$ after each move.

a) A translation 5 units right and 2 units down

b) A translation 2 units left and 3 units up

c) A reflection in the mirror line that goes straight up through 5 on the x-axis

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Transformations

Name _____ Date _____ No calculator Version 2

Scan to practise this topic online

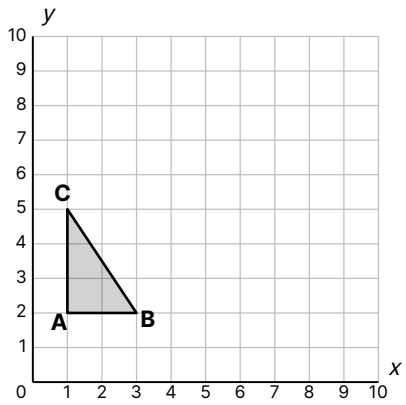
KEY WORDS translation *(nekehanga)* · reflection *(whakaata)* · rotation *(hurihanga)* · mirror line *(rārangi whakaata)*

A Translations ●○○

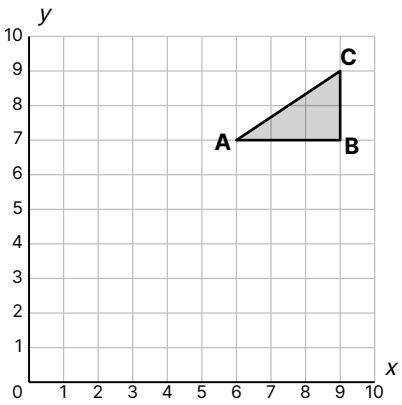
Remember: A translation slides a shape. Every corner moves the same distance in the same direction. Label the corners of the image A′, B′, C′.

1. Draw the image of each shape.

a) Translate the shape 6 units right and 3 units up.

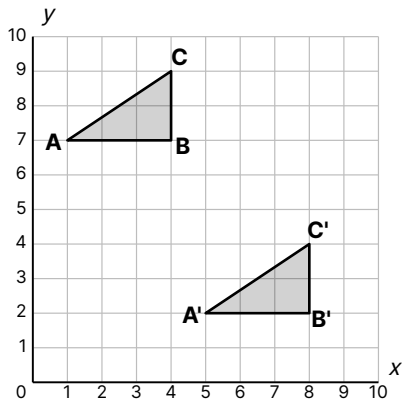


b) Translate the shape 4 units left and 5 units down.



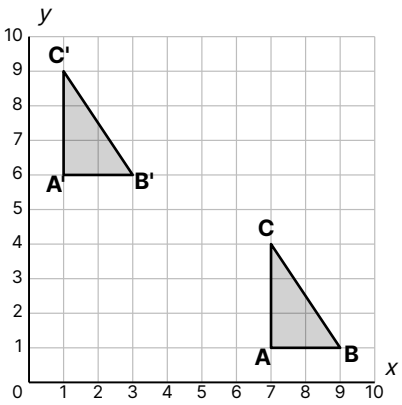
2. Shape ABC has been translated to A′B′C′. Describe the translation. Say how many units it has moved right or left, and up or down.

a)



_____

b)



_____

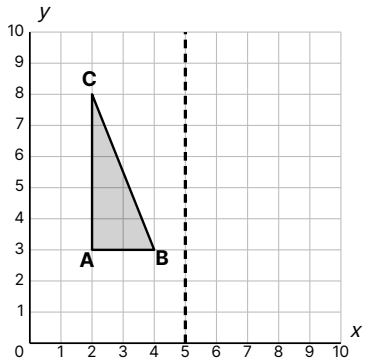
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B Reflections ●●○

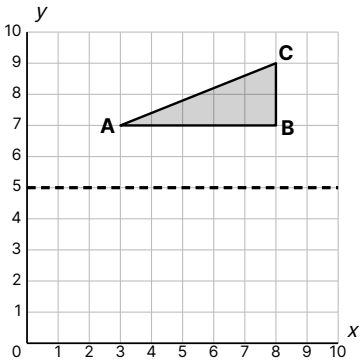
Remember: A reflection flips a shape over a mirror line. Each corner and its image are the same distance from the mirror line, on opposite sides of it.

3. Draw the image of each shape.

a) Reflect the shape in the dashed mirror line.



b) Reflect the shape in the dashed mirror line.

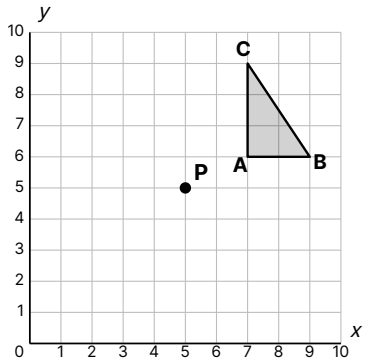


C Rotations ●●●

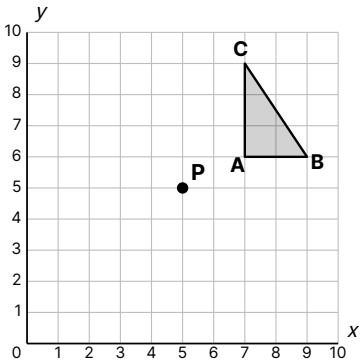
Remember: A rotation turns a shape about a point. Tracing paper helps: hold the point P still and turn the paper.

4. Draw the image of each shape.

a) Rotate the shape a half turn (180°) about the point P.



b) Rotate the shape a quarter turn (90°) anticlockwise about the point P.



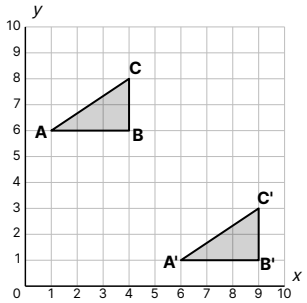
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D Naming the transformation ●●○

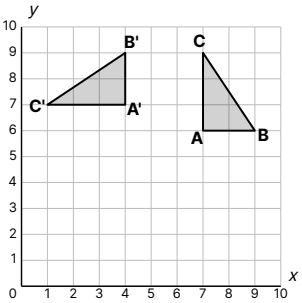
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5. Shape ABC has been moved to A′B′C′. Is each move a translation, a reflection or a rotation?

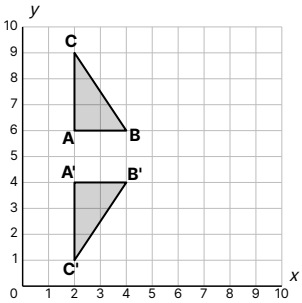
a)



b)



c)



6. Which kind of transformation is being described?

a) A shape is turned upside down by a half turn about a point. _____

b) A shape slides 3 units up. _____

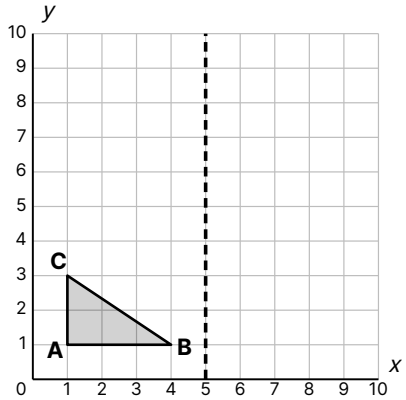
c) A shape and its image are the same distance from a line, on opposite sides of it. _____

E Two-step transformations ●●●

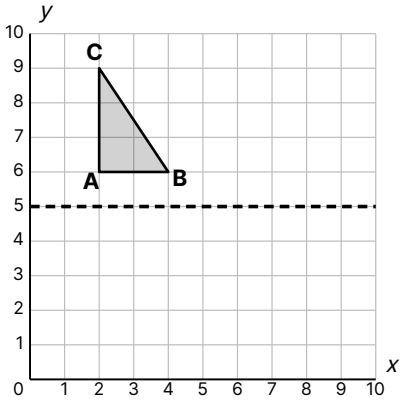
Watch out: Do one step at a time. Draw the shape after the first step lightly, then move that new shape for the second step.

7. Draw the final image of each shape.

a) Translate the shape 5 units up. Then reflect the new shape in the dashed mirror line.



b) Reflect the shape in the dashed mirror line. Then translate the new shape 4 units right.



Extension

8. Without drawing, write the coordinates of the image of the point $(2,\ 6)$ after each move.

a) A translation 5 units right and 2 units down

b) A translation 2 units left and 3 units up

c) A reflection in the mirror line that goes straight up through 5 on the x-axis

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